



Infrastructure Governance Manual

This technical manual serves as the definitive engineering blueprint for the ShineGuard Hulo platform. All system operations are governed by the deterministic thresholds and forensic security protocols documented herein.

Master Governance Matrix

Category	Metric	Active Rule	Thesis-Level Rationale	
Physics		Voltage	< 180V	Prevents current spikes in LED driver SMPS per IEC 60038 standards.
Hardware		Thermal	> 85°C	Standard industrial semiconductor redline to prevent silicon gate aging.
Safety		Current	> 5.0A	High-speed overload protection for industrial lighting circuitry.
Audit		Integrity	HMAC-256	Blockchain-style chaining to ensure logs are immutable and forensic.

Verification Note: This matrix defines the "Protective Shield" of the platform. Any data point exceeding these values triggers an immediate autonomous safety escalation and real-time dashboard notification.

Telemetry Hardening & Logic

1. Electrical & Performance Suite

Metric	Nominal Range	Warning Level	Critical Trip	Compliance
Voltage	210V - 240V	< 200V / > 250V	< 180V	IEC 60038
Temperature	20°C - 55°C	> 65°C	> 85°C	Industrial Grade
Current (A)	0.01A - 2.0A	> 2.5A	> 5.0A	IEC 60598-1

Engineering Analysis (Electrical): Following IEC 60598-1 regulations, ShineGuard implements a 5.0A hard-trip to prevent cable combustion. The voltage floor of 180V is calculated to protect against Switch-Mode Power Supply (SMPS) oscillation during brownouts, which is a primary cause of LED failures in urban deployments.

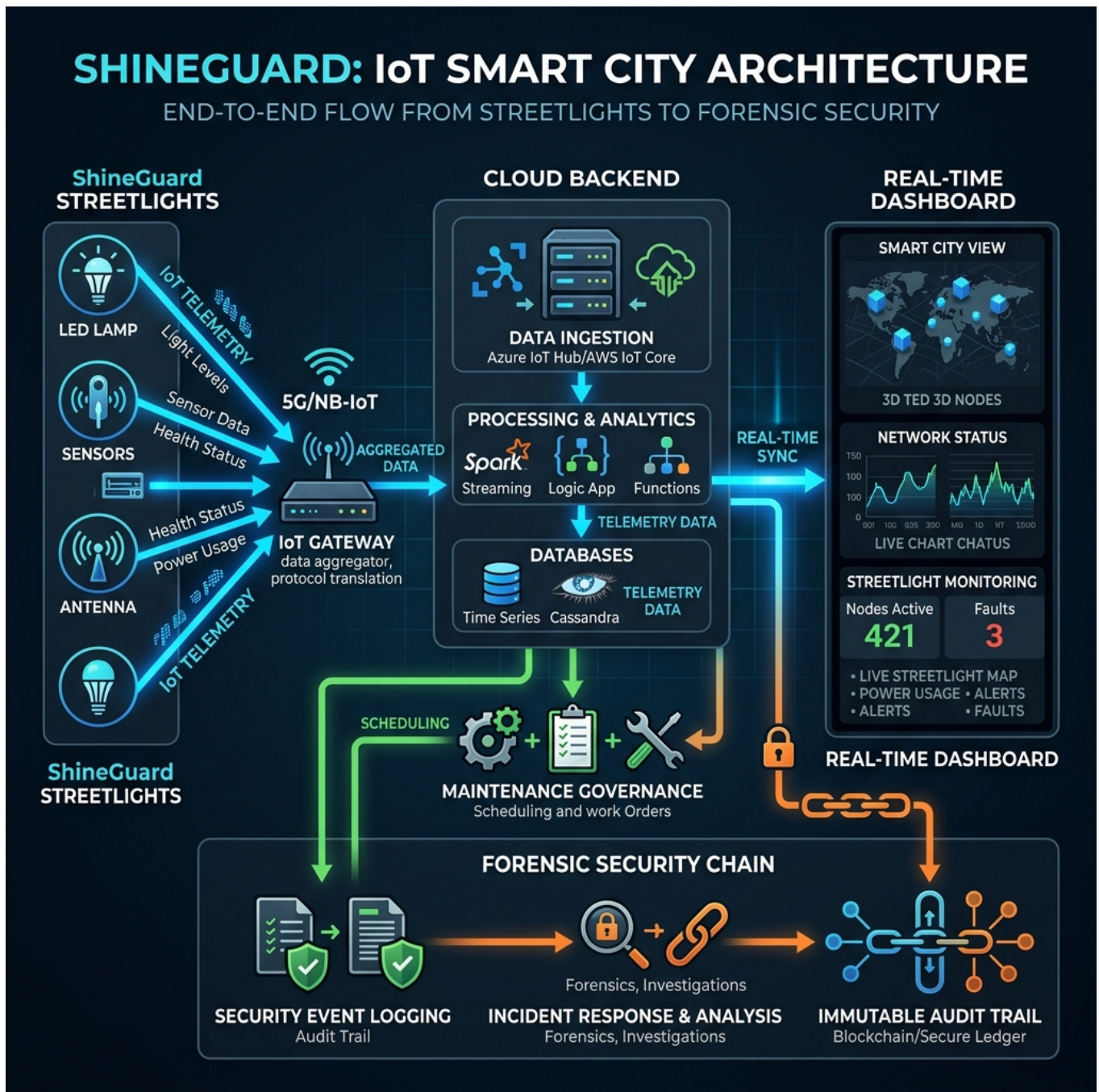
2. Environmental Telemetry Suite

Metric	Nominal Range	Warning Level	Compliance Standard
Ambient Light	10 - 1000 lux	< 10 lux	EN 13201 (Urban)
Humidity	10% - 60%	> 85% RH	IP65 Enclosure Code

Engineering Analysis (Atmospheric): Urban road safety under EN 13201 mandates specific illumination levels for pedestrian health. ShineGuard uses a 10 lux floor to trigger emergency override lighting. Humidity tracking predicts gasket failure in IP65 housings before water ingress damages the circuitry.

System Architecture & Connection Map

The visual schematic below documents the end-to-end data journey from physical streetlight sensors to the forensic archival vault.



Inter-Section Connectivity Legend:

Infrastructure Layer:

Physical sensor array for real-time electrical monitoring.

Core Intelligence:

Cloud-based deterministic processing and rule enforcement.

User Experience:

Premium dashboard providing human-in-the-loop oversight via Firebase.

Forensic Security:

Immutable audit trail providing legal accountability for every action.

Forensic & Security Manifesto

1. Defensive Layer: Active Mitigation

Brute-Force Lockdown (15-Min Policy):

The system tracks failed login vectors. Upon 5 consecutive failures, the IP is globally blacklisted for 15 minutes to prevent automated dictionary attacks.

Secure Session Authorization (SBA):

Critical bulk operations are gated by a 5-minute administrative window, requiring fresh re-authentication for state-changing commands.

2. Defensive Layer: Data Integrity

HMAC-SHA256 Forensic Chaining:

Each audit log entry is mathematically "Linked" to the previous entry, creating an immutable sequence. This creates a blockchain-style proof of non-repudiation.

Off-System Audit Validation:

The system provides a dedicated Forensic Verification Link (`tools/verify_logs.php`) allowing third-party auditors to verify the integrity of the log chain.

3. Defensive Layer: Future-Proofing

MFA Architectural Readiness:

The Identity Service includes predefined hooks and logic for TOTP (Google Authenticator) Multi-Factor Authentication for future military-grade hardening.